

Fraud Transaction Analysis

Exploratory analysis, hypothesis testing, post-ANOVA follow-up and model benchmark

Business question

Which variables best identify fraud risk, and how well can the variables support fraud classification?

10,000

transactions

500

fraud cases

Key conclusion

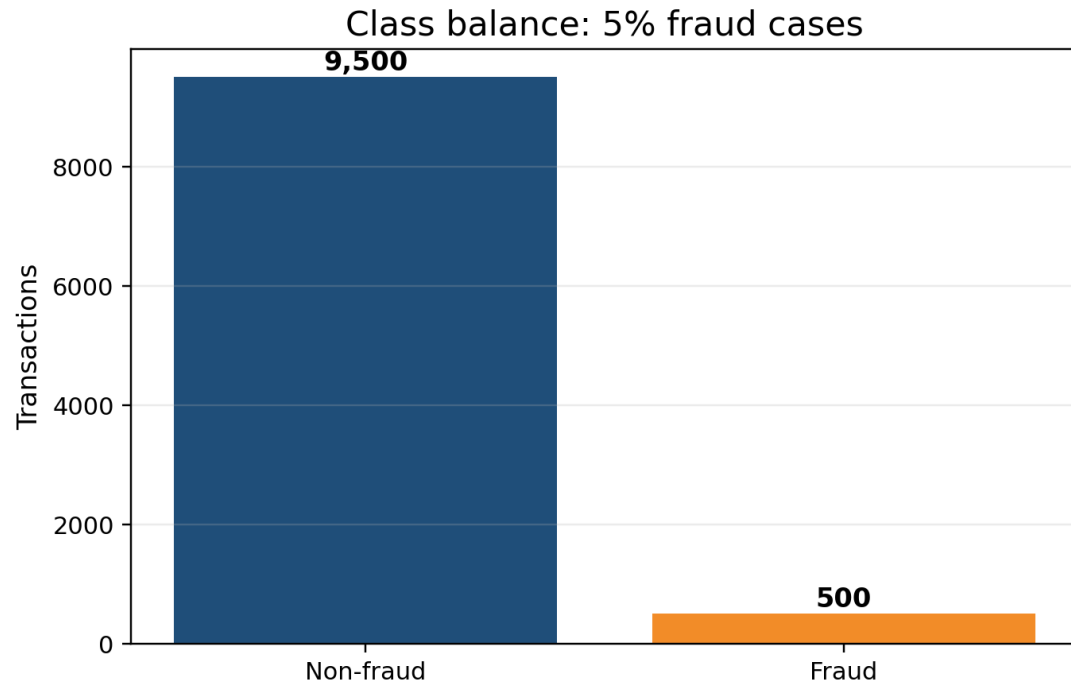
High device/IP risk scores, NG country and Night transactions are the strongest indicators in this dataset.

5.0%

fraud rate

Dataset overview

Target variable and data fields used in the analysis



Data fields used

Target: is_fraud (Fraud / Non-fraud)

Numeric fields: amount, log amount, device risk score, IP risk score

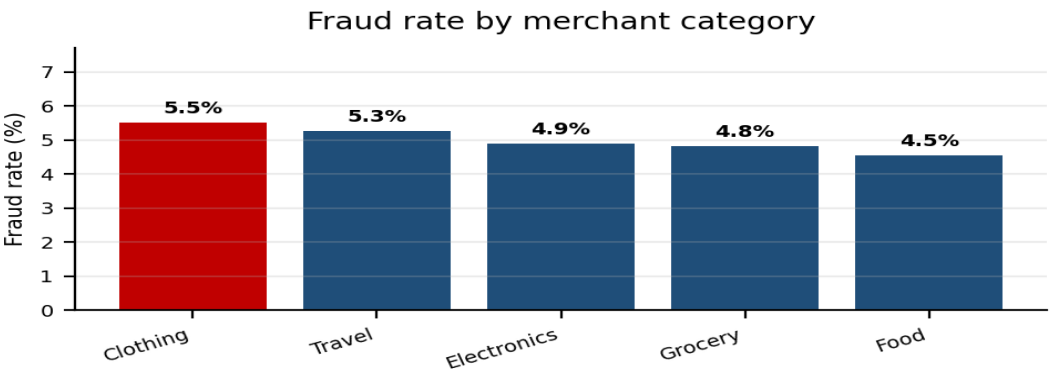
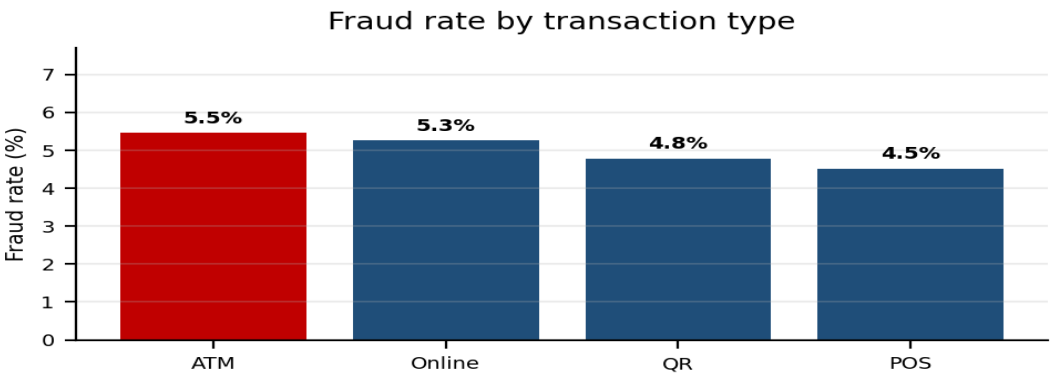
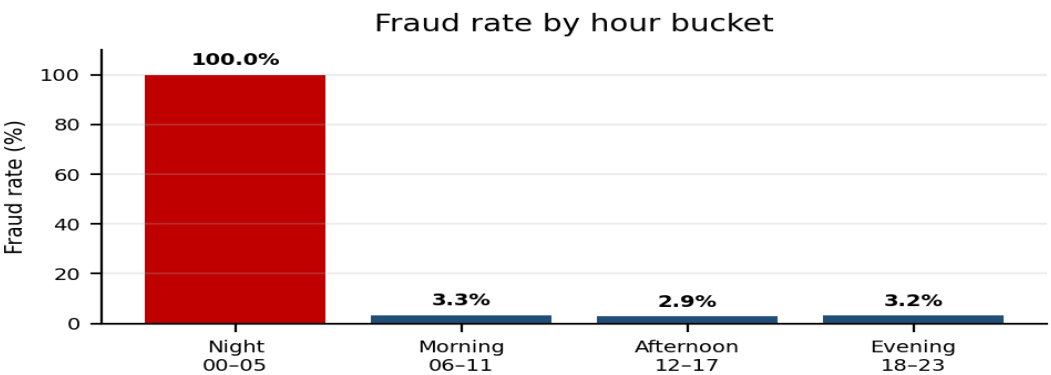
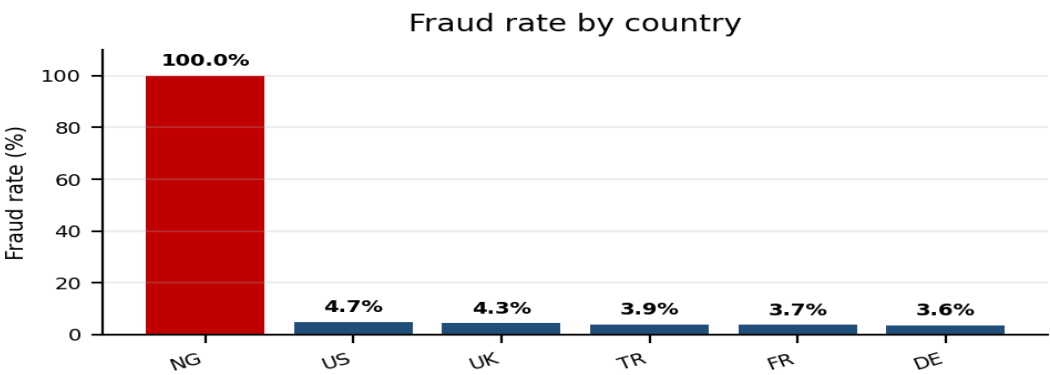
Categorical fields: country, hour bucket, transaction type, merchant category

Fraud is imbalanced, so precision, recall and fraud capture rate are more useful than accuracy alone.

The dataset contains 10,000 transactions, of which 500 are fraud. This creates a realistic fraud-monitoring problem: the risky cases are a small share of total activity.

Categorical fraud patterns

Fraud rate comparison across country, time period, transaction type and merchant category



Main finding: NG and Night (00-05) have the highest fraud rates in this synthetic dataset. Transaction type and merchant category show much smaller differences, so they are weaker standalone indicators.

Chi-square test: categorical variables and fraud

Tests whether fraud status is associated with each categorical field

Hypotheses used for each chi-square test

H0: Fraud status is independent of the category.
H1: Fraud status is associated with the category.

Why this test was appropriate

Both variables are categorical: fraud status is Fraud / Non-fraud, and the tested fields are country, hour bucket, transaction type and merchant category.

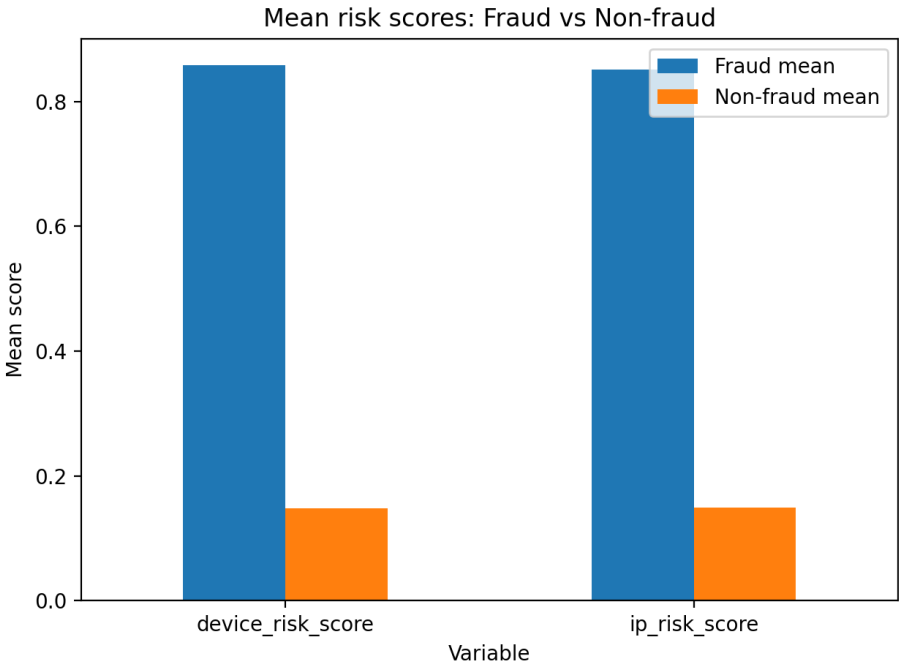
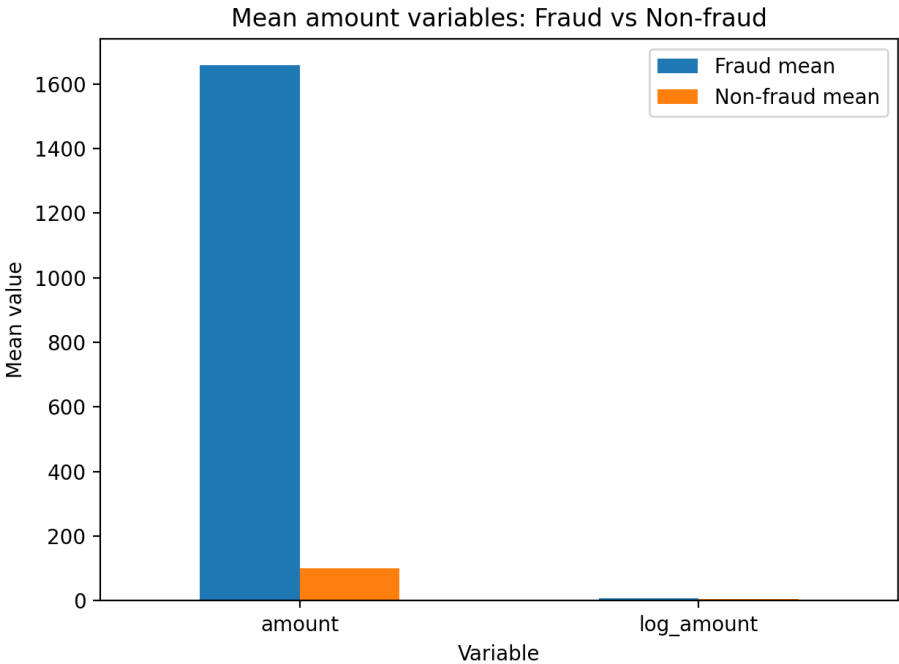
Variable tested	p-value	Decision	Interpretation
Country	< 0.001	Significant	Fraud differs by country; NG has the highest fraud rate.
Hour bucket	< 0.001	Significant	Fraud differs by time period; Night (00–05) is highest.
Transaction type	0.401	Not significant	No strong evidence that fraud differs by transaction type.
Merchant category	0.663	Not significant	No strong evidence that fraud differs by merchant category.

Finding

Country and hour bucket are statistically associated with fraud. Transaction type and merchant category are weaker categorical indicators in this dataset, so they should not be used alone for fraud triage.

T-test findings: fraud vs non-fraud means

Welch two-sample t-tests compare numeric variables between Fraud and Non-fraud groups



Purpose: the t-tests check whether fraud and non-fraud transactions have different average numeric values. Significant results mean the variable helps separate suspicious transactions from normal transactions.

Finding: fraud transactions have much higher mean amount, log amount, device risk score and IP risk score (all $p < 0.001$). Device and IP risk scores show the clearest separation.

ANOVA: p-value findings for risk-score differences

Focused ANOVA tests for device/IP risk scores across country and hour bucket

Focused ANOVA p-value findings

All tests compare mean risk scores across country or hour-bucket groups.

Grouping variable	Numeric variable	p-value	Conclusion
Country	Device risk score	< 0.001	Significant
Country	IP risk score	< 0.001	Significant
Hour bucket	Device risk score	< 0.001	Significant
Hour bucket	IP risk score	< 0.001	Significant

ANOVA tests whether the average risk score is the same across all countries or all hour buckets. A significant p-value means at least one group has a different average risk score.

What the ANOVA results say

All four focused ANOVA tests are statistically significant ($p < 0.001$).

Device and IP risk scores are not evenly distributed across countries.

Device and IP risk scores also differ by time period.

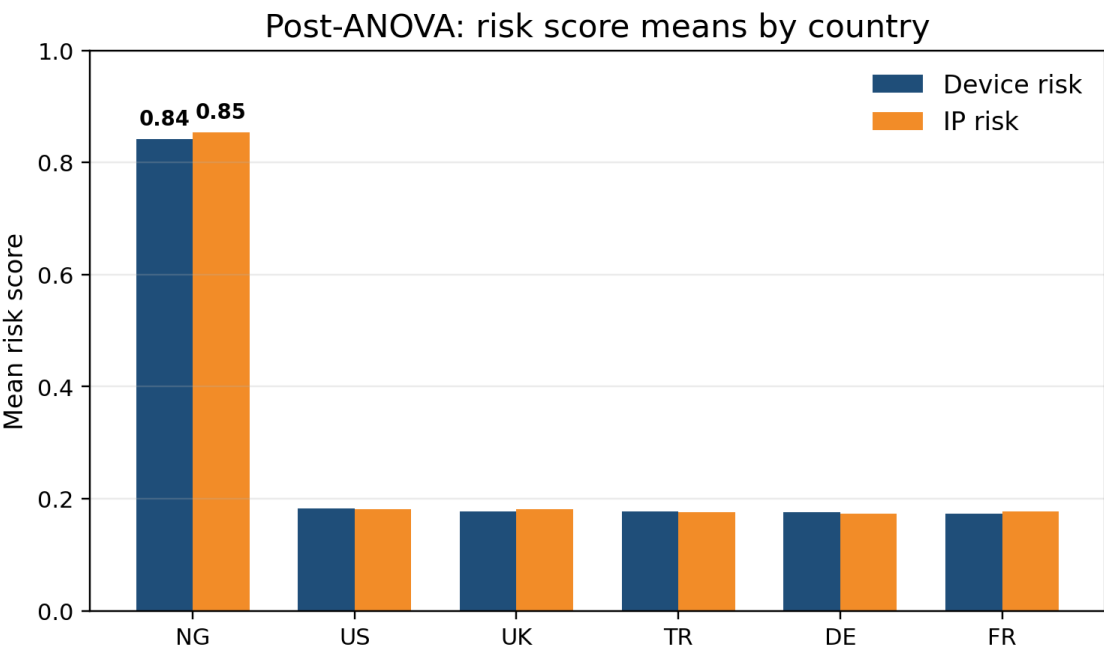
This supports country and hour bucket as useful risk context, alongside the risk-score variables.

Why follow-up testing is needed

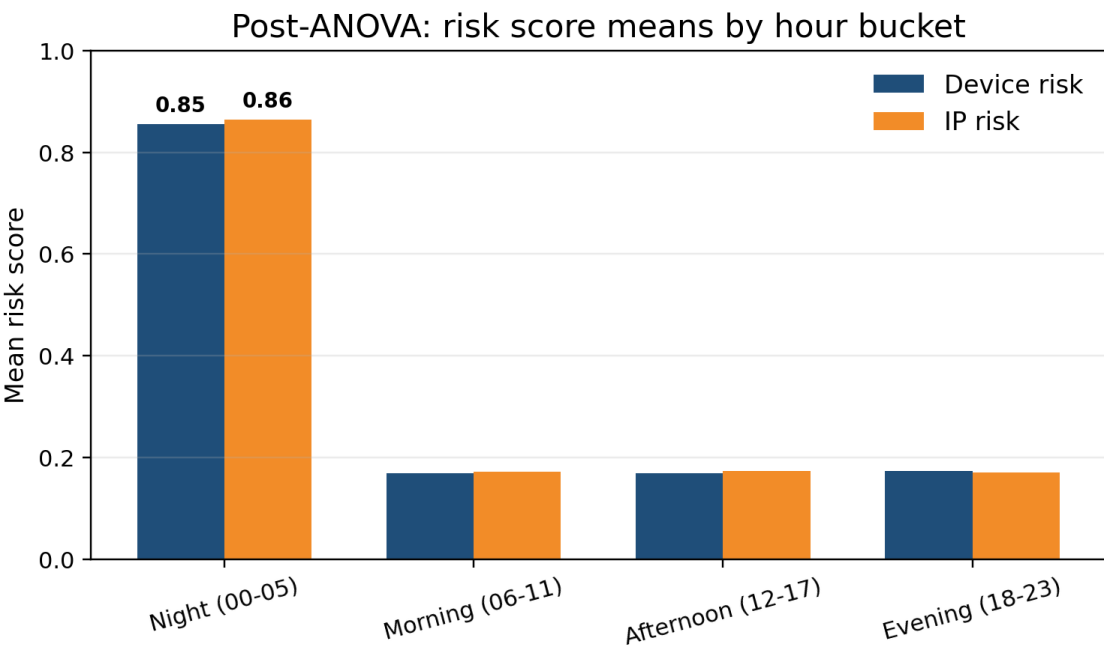
ANOVA only confirms that a difference exists somewhere. It does not identify which country or hour bucket is different. The next slide uses post-ANOVA follow-up testing to identify the groups driving the result.

Post-ANOVA follow-up: which groups are different?

Targeted Welch t-tests confirm the main high-risk groups after ANOVA



Country: NG has significantly higher mean device and IP risk scores than every other country. Other country pairs are not meaningfully different.



Hour bucket: Night (00–05) has significantly higher mean device and IP risk scores than Morning, Afternoon and Evening. Other time buckets are not meaningfully different.

Targeted follow-up tests

NG and Night are statistically higher-risk groups for device/IP scores

Comparison	Variable	High-risk mean	Other mean	Result
NG vs all other countries	Device risk score	0.841	0.177	p < 0.001
NG vs all other countries	IP risk score	0.854	0.178	p < 0.001
Night vs all other hour buckets	Device risk score	0.855	0.171	p < 0.001
Night vs all other hour buckets	IP risk score	0.864	0.171	p < 0.001

Business interpretation

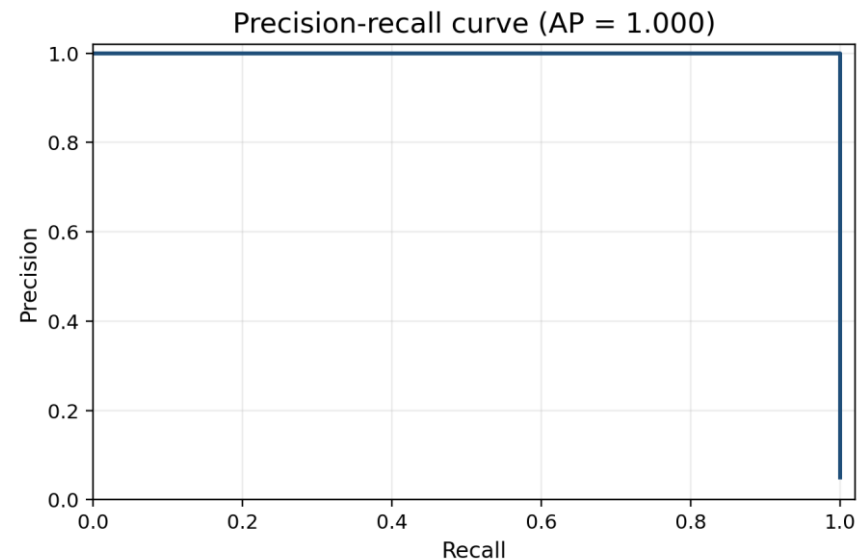
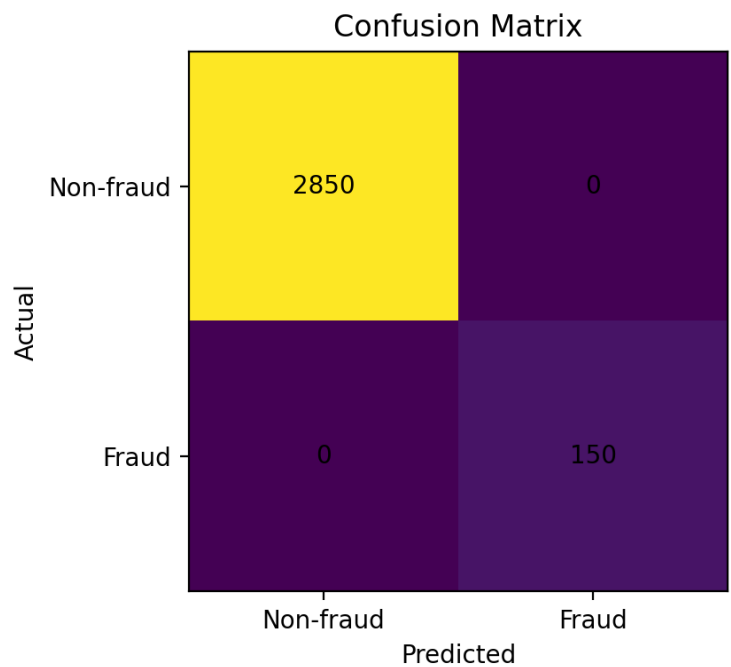
NG and Night are not just statistically significant; their average risk scores are around five times higher than other groups.

These groups should be used as secondary triage factors together with device/IP risk scores, not as standalone rules.

This approach supports prioritising limited investigator time toward transactions with the strongest risk evidence.

Predictive model benchmark

Random Forest used to test whether the variables can predict fraud



Model result

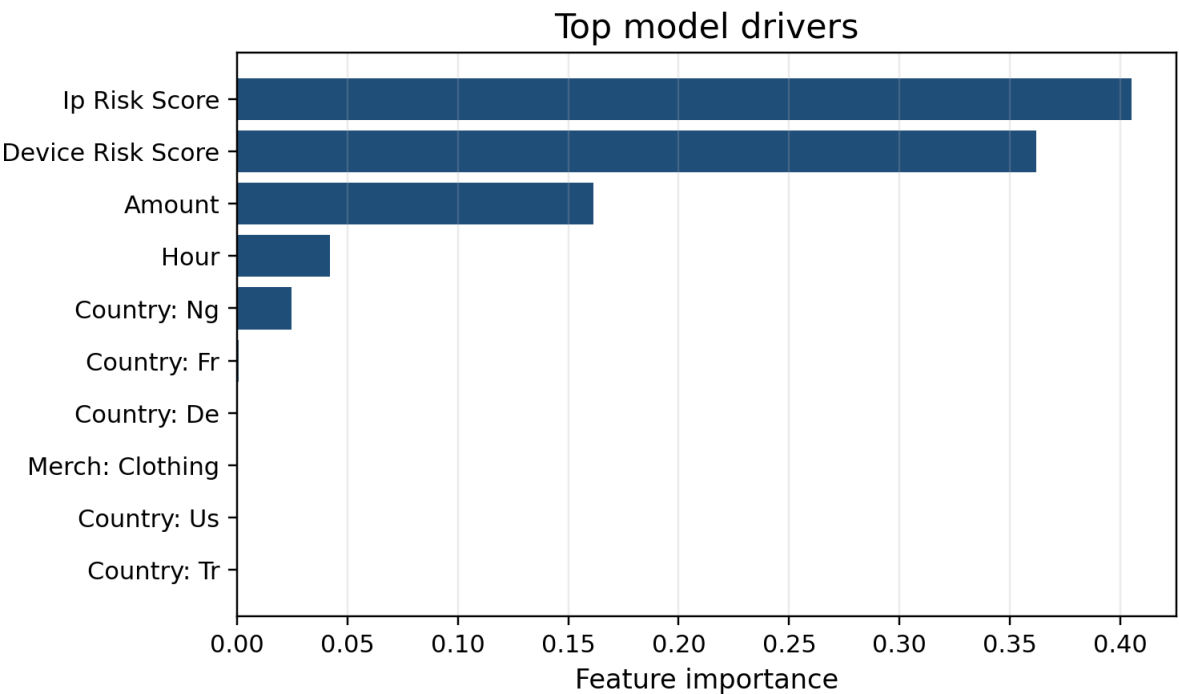
The model separates fraud and non-fraud perfectly on the test split: precision = 1.00 and recall = 1.00.

Interpretation

This confirms that the synthetic data contains strong fraud signals, mainly risk scores and high-risk segments. Real-world deployment would need leakage checks and validation on future periods.

Model drivers align with statistical findings

Feature importance supports the hypothesis test results



Consistent evidence

Device risk score and IP risk score are the strongest numeric indicators.

Amount also helps separate fraud from non-fraud.

Country and hour bucket provide useful context, especially NG and Night.

Transaction type and merchant category are weak in this dataset.

The statistical tests and model are telling the same story: risk-score fields are the primary fraud indicators, while country and time help prioritise suspicious cases.

Final findings and recommendations

Main findings

- Fraud is strongly associated with country and hour bucket.
- Fraud transactions have higher amount, device risk score and IP risk score.
- Post-ANOVA follow-up shows NG and Night drive the risk-score differences.
- Transaction type and merchant category are weaker indicators.

Limitations

The dataset is synthetic and has very clean signals. A real fraud dataset would likely have noisier patterns, changing fraud behaviour and more false positives.

Recommendations

- Use high device/IP risk scores as the strongest fraud indicators.
- Use country, hour bucket and amount as supporting explanatory variables.
- Summarize findings using clear charts and statistical tests for reporting.
- Validate model performance on future data before operational use.

Next step

Repeat the analysis on new time periods and monitor precision/recall drift over time.